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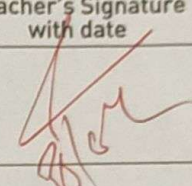
Subject: IDS Roll No.: 29 Exp. No.: _____

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Performed on: _____

Submitted on: _____

Marks
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Teacher's Signature with date


1] Define an algorithm. List any 4 characteristics of an algorithm

→ An algorithm is a finite sequence of well-defined step-by-step instruction used to solve a specific problem or perform a computation

Characteristics :-

1) Input :- An algorithm takes zero or more inputs

2) output :- It produces at least one output

3) Definiteness :- Each step is clear, precise unambiguous.

4) Finiteness :- The algorithm terminates after a finite no of steps.

2] Define data structure & Abstract Data Type (ADT) give one example of each

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐

Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐



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Q2 Define data structure and Abstract data type (ADT) give one example of each.

A data structure is a way of organizing, storing and managing data in memory as that it can be accessed and modified efficiently.

An Abstract data Type (ADT) is a logical description of a data type that specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples :

- Data structure Array
- ADT : stack (operations : push, pop, peek)

Q3. Explain the process of converting a problem into a program. illustrate the role of algorithm in this process.

1. Problem Definition : understand requirements and constraints.
2. Analysis : identify inputs, outputs and processing.
3. Algorithm design : create step-by-step logic to solve the problem.
4. Flowchart/Pseudocode : Represent the algorithm visually or textually.

5. Coding : convert algorithm into a program using a programming language.
6. Testing & Debugging : check correctness. and remove errors.
7. Documentation & maintenance : improve and maintain the program.

Role of Algorithm.

The Algorithm acts as a blueprint of the solution. It ensures correctness, efficiency and makes coding easier and error free.

Q4. Explain linear and non-linear data structures with suitable examples.

Linear data structure :

Data elements are arranged sequentially.

- Array
- Stack
- Queue
- Linked list

Non-linear Data structure :

Data elements are arranged in a hierarchical or interconnected manner.

Example :
• Tree
• Graph

Q5. Differentiate between algorithm and program. Analyze their roles in SDLC.

Algorithm	Program
Step-by-step logic	written code
Language independent	Language dependent
Design level	Implementation level
No syntax rules	follows syntax rules

Role in SDLC :

- Algorithm : used in design phase to plan the solution.
- Program : used in implementation phase to execute the solution on a computer.

Q6. Examine the relationship among data, data structure and algorithm. Justify how the choice of data structure affects algorithm efficiently.

Relationship :

- Data is raw information
- Data structure organizes the data
- Algorithm processes the data using data structures

Justification :

The efficiency of an algorithm depends on the data structure used.

- Searching in an array $\rightarrow$ slower ($O(n)$)
- Searching in a binary search tree $\rightarrow$ faster ($O(\log n)$)

Conclusion :

choosing the right data structure improves time and space efficiency of Algorithms.